Introduction to Internet of Things

Assignment-Week 5

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

When different processing logic are applied to the same IoT networked devices or applications, it gives rise to?

a. Large scale of co-operation

b. Semantic conflicts

c. Both (a) and (b)

d. Neither (a) nor (b)

Correct Answer: b. Semantic conflicts

Detailed Solution: When different processing logic are applied to the same IoT networked devices or applications, it gives rise to Semantic conflicts.

See lecture 21 (Interoperability in Internet of Things) @ 03:41.

QUESTION 2:

State True or False.

Statement: "Interoperability is a characteristic of a product or system."

a. True
b. False

Correct Answer: a. True

Detailed Solution: Interoperability is a characteristic of a product or system.

Refer Lecture 21@5:51

QUESTION 3:

Interoperability is important in the context of IoT so that -

a. Physical objects remain isolated
b. Networks remain isolated
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: d. Neither (a) nor (b)

Detailed Solution: Interoperability is important in the context of IoT so that –

a. Physical objects can interact
b. Any device can communicate with any other device
c. Networks integrate seamlessly

Refer Lecture 21@07:34

QUESTION 4:

State whether the following statement is true or false

Statement: "Use of different programming languages such as JavaScript, Python, JAVA, and others is an example of heterogeneity in IoT. This brings in the need for interoperability."

a. False

b. True

Correct Answer: b. True

Detailed Solution: Use of different programming languages such as JavaScript, Python, JAVA, and others is an example of heterogeneity in IoT. This brings in the need for interoperability

(Please refer Lecture 21@09:12)

QUESTION 5:

State True or False.

Statement: "The interoperability between devices and device users in terms of message formats is called Syntactic Interoperability."

a. True

b. False

Correct Answer: a. True

Detailed Solution: The interoperability between devices and device users in terms of message formats is called Syntactic Interoperability.

Refer Lecture 21@17:06.

QUESTION 6:

What is the full form of UMB in IoT interoperability?

a. Universal Middleware Bridge

b. Universal Main Bridge

c. Universal Main Bracket

d. None of these

Correct Answer: a. Universal Middleware Bridge

Detailed Solution: UMB stands for Universal Middleware Bridge.

Refer Lecture 21@22:16.

QUESTION 7:

Which of the following pins is NULL in the DHT-11 sensor?

a. Pin 2

b. Pin 4

c. Both (a) and (b)

d. Neither (a) nor (b)

Correct Answer: d. Neither (a) nor (b)

Detailed Solution: Pin 3 is NULL in the DHT-11 sensor.

Refer Lecture 24@03:26.

QUESTION 8:

Arduino MEGA microcontroller board is also known as?

a. MEGA 3370

b. MEGA 2570

c. MEGA 9980

d. None of these

Correct Answer: d. None of these

Detailed Solution: Arduino MEGA microcontroller board is also known as MEGA 2560.

Refer Lecture 24@03:43.

QUESTION 9:

A _____ provides high-precision rotary motion of 0 to 180 degrees.

a. Solenoid Valve

b. Relay

c. Servo Motor

d. None of these

Correct Answer: c. Servo Motor

Detailed Solution: A Servo Motor provides high-precision rotary motion of o to 180 degrees.

Refer Lecture 25@04:47.

QUESTION 10:

What does the following code snippet do in interfacing a servo motor with the Arduino MEGA board?

int servoPin = 12;

a. Declares pin for connecting servo motor

b. Declares pin for providing power to MEGA board

c. Declares pin for Ground supply for servo motor

d. None of these

Correct Answer: a. Declares pin for connecting servo motor

Detailed Solution: *int servoPin = 12;* Declares pin for connecting servo motor.

Refer Lecture 25@07:54.

QUESTION 11:

What does the *delay(1000);* line do mean in an Arduino sketch?

a. It introduces a delay of 1 second

b. It introduces a delay of 10 seconds

c. It introduces a delay of 100 milliseconds

d. None of these

Correct Answer: a. It introduces a delay of 1 second

Detailed Solution: The line *delay(1000);* in an Arduino sketch introduces a delay of 1 second.

Refer Lecture 25@08:25

QUESTION 12:

State True or False.

Statement: "Additional hardware is required to load a program into the Arduino Uno board using a computer."

a. True
b. False

Correct Answer: b. False

Detailed Solution: No additional hardware is required to load a program into the Arduino Uno board using a computer.

Refer Lecture 22@05:11

QUESTION 13:

Arduino Uno board has _____ number of PWM pins?

a. 9
b. 4
c. 8
d. None of these

Correct Answer: d. None of these

Detailed Solution: Arduino Uno board has 6 number of PWM pins.

Refer Lecture 22 @ 07:13.

QUESTION 14:

Consider the following Arduino sketch.

```
int r = 4;
int g = 5;
void setup(){

Serial.begin(9600);
pinMode(r,INPUT);
pinMode(g,__??__);
}

void loop()
{
  int val = analogRead(r);
  if( val > 10)
  {
      digitalWrite(g,HIGH);
  }
  else
  {
      digitalWrite(g,HIGH);
  }
  delay(500);
}
```

What must be inserted in the place of '??' within the second pinMode() function in void setup()?

a. INPUT

b. OUTPUT

c. None of these

d. Anyone of these is okay

Correct Answer: b. OUTPUT

Detailed Solution: Since pin 'g' is writing digital values, it is required to set in OUTPUT mode. Refer Lecture 22-23.

QUESTION 15:

State True or False.

Statement: "The "Verify" option in the Arduino IDE checks the code for compilation errors."

a. True
b. False

Correct Answer: a. True

Detailed Solution: The "Verify" option in the Arduino IDE checks the code for compilation errors.

Refer Lecture 24.

**********END**********